Speech Recognition (DSAI 456)

Lecture 2: Foundations

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Agenda

Objectives: by the end of this lecture you should be able to name the parts of a syllable, read amplitude/frequency/period off a waveform, explain what sampling and quantization actually do to a signal, distinguish pitch/loudness/intensity/F0, and read formants off a spectrogram.

- **Linguistic units**: syllable, vowel, consonant
- **Acoustics**: waveform, amplitude, frequency, Hertz, period
- **Digitization**: sampling, Nyquist and aliasing, quantization, framing
- **Describing the signal**: pitch, loudness, intensity, F0
- **Formants**: why vowels sound different
- **Spectrogram**: putting all of it into one picture

Recap: Last Week

- A speech signal is 16,000 numbers a second with no boundaries marked between words.
- WER = (S + D + I) / N. It has no ceiling and it hides which errors matter.
- This week: what *is* that stream of numbers, physically and linguistically, before any model sees it?

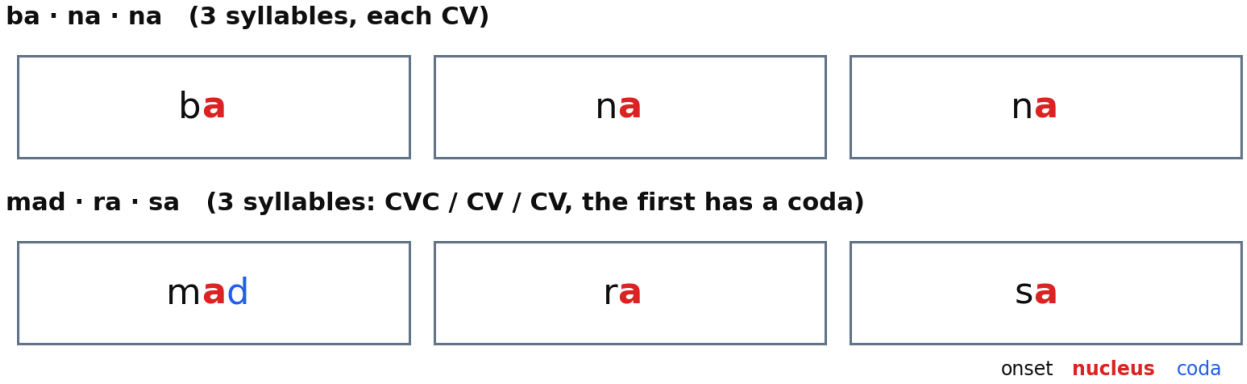
Part 1

What Are We Trying to Recognize?

Syllables, vowels, consonants

Syllable = Onset + Nucleus + Coda

- A **syllable** is a single unbroken unit of spoken sound built around exactly **one nucleus**, usually a vowel, with or without surrounding consonant sounds.
- The **nucleus** is the core (every syllable has exactly one); consonants before it are the **onset**, after it the **coda**. Nucleus + coda = the **rime**. The nucleus can also be a consonant: *button*.
- Examples: "banana" and مَدْرَسَة (madrasa).



Vowel vs. Consonant

Vowels

- Open vocal tract, no major obstruction; normally voiced (exceptions: whispered vowels, and devoiced vowels such as the final *u* in Japanese *desu*).
- Carry the syllable nucleus; tend to be louder and longer.

Important
Every language uses this vowel/consonant split, but how do I tell them apart on a waveform?

Consonants

- Some constriction of the airflow (lips, tongue, teeth).
- Voiced (/b/, /n/) or voiceless (/p/, /s/); plosive, fricative, or nasal.

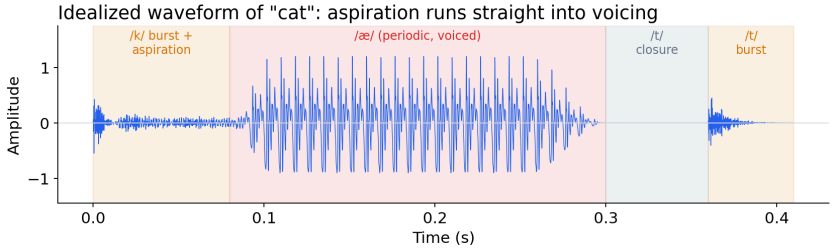
Part 2

What Sound Physically Is

Waveform, amplitude, frequency, Hertz, period

The Waveform

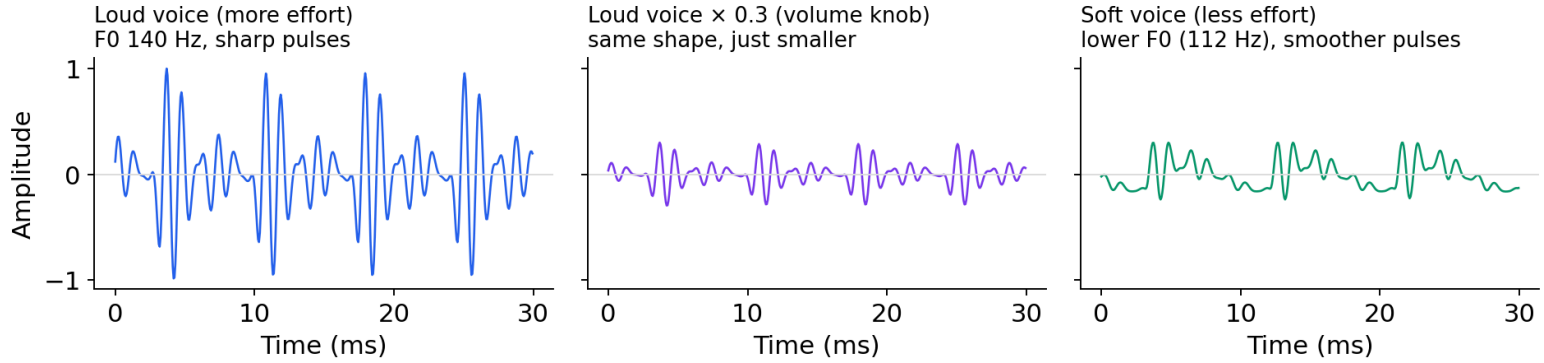
- Speech is air pressure changing over time. The **waveform** plots that pressure (y-axis) against time (x-axis).
- A pure tone is a sine wave: $y = A\sin(2\pi ft)$
- **Voiced** speech (vowels, nasals) is a sum of many such waves at once: messy-looking, but not random, because it repeats.
- **Voiceless** sounds (/s/, /f/, bursts, aspiration) are aperiodic noise: they really are random-looking.
- Silence, a vowel, and a consonant each leave a visually distinct signature.



Idealized waveform for **"cat"**: the /k/ burst and aspiration (aperiodic noise) run straight into the periodic, voiced /æ/; then silence while the tongue closes for /t/, and its release burst.

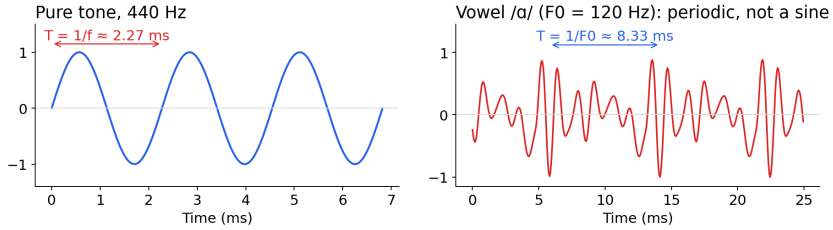
Amplitude

- **Amplitude** is the size of the pressure swing — how far the waveform deflects from zero.
- Larger amplitude → louder sound (we'll formalize "loud" as intensity shortly).
- Turning a **volume knob** only rescales the waveform: same *shape*, different *size*.



Frequency, Hertz, Period

- **Frequency** (f): how many cycles of the wave happen per second.
- **Hertz (Hz)**: the unit of frequency — 1 Hz = 1 cycle/second.
- **Period** (T): the time for *one* cycle to complete: $T = 1/f$.
- A pure 440 Hz tone repeats every $1/440 \approx 2.27$ ms.
- Voiced speech is **not** a pure tone: it's quasi-periodic, a fundamental frequency plus harmonics at integer multiples of it. Voiceless sounds (/s/, /f/) have no period at all.



Part 3

From Continuous Sound to Numbers

Sampling, quantization, framing

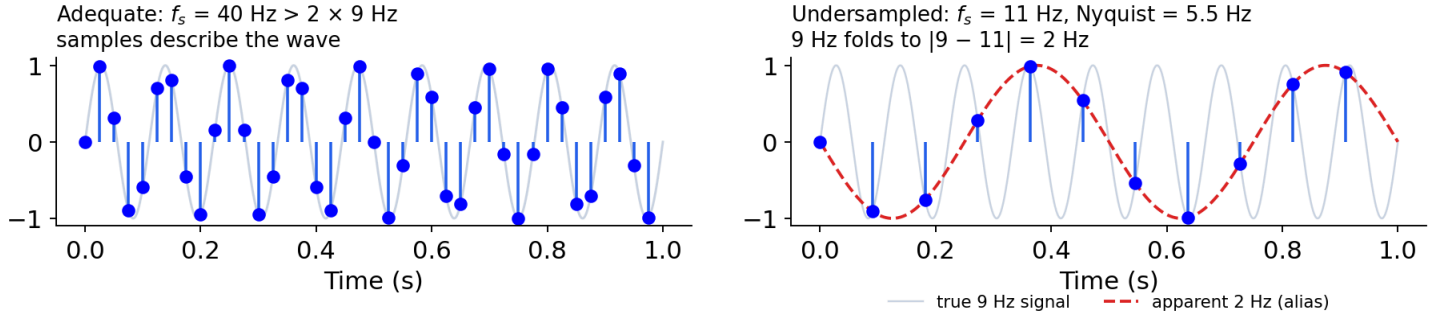
Sampling

- The waveform is **continuous** — infinitely many values in any interval. A computer can't store that, so **sampling** measures the amplitude at regular intervals and keeps only those points.
- **Sampling rate** f_s: samples/second (Hz).
  - Phones and laptops typically record at **44.1 or 48 kHz**.
  - Most ASR models expect **16 kHz** (wideband speech).
  - Telephone speech: **8 kHz** (narrowband, roughly 300–3400 Hz passes).
- A sampling rate can only represent frequencies below **half** of it (next slide).

> **Important**
> 16 kHz keeps everything below 8 kHz, which covers the frequencies that matter most for intelligibility. That's why it's the ASR standard.

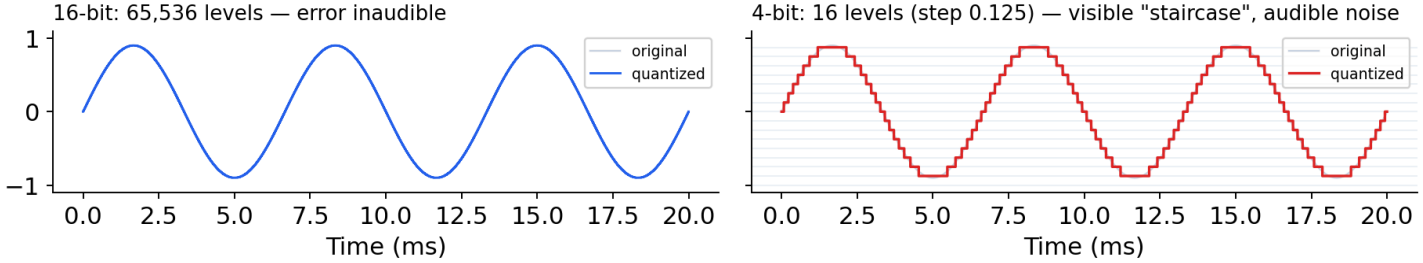
Nyquist and Aliasing

- **Nyquist rule**: to represent a frequency f_{max} you need $f_s > 2f_{max}$. Equivalently, $f_s/2$ (the **Nyquist frequency**) is the highest frequency a sampling rate can hold.
- A component above $f_s/2$ doesn't disappear: it **folds** to $|f - k \cdot f_s|$ (the nearest multiple k of f_s), a false lower-frequency wave. That is **aliasing**.
- In the figure: $f_s = 11$ Hz, so Nyquist is 5.5 Hz, and the 9 Hz wave shows up as $|9 - 11| = 2$ Hz.



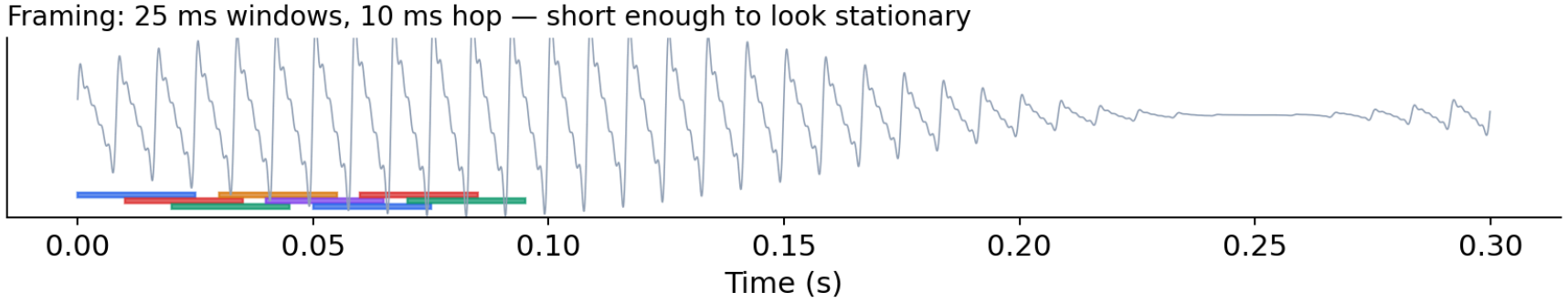
Quantization

- Bit: the basic unit of digital storage — a single binary digit, either 0 or 1. n bits can represent 2^n distinct values, so more bits = more possible amplitude levels to round to.
- Bit depth is simply how many bits are used per sample.
- **Quantization** rounds each sample's amplitude to the nearest of a fixed number of levels, set by the **bit depth**.
- 16-bit audio → 65,536 levels, rounding error (**quantization noise**) far too small to hear.
- Fewer bits → fewer levels → a visible "staircase" and audible noise.



Framing

- Speech is **non-stationary** — its frequency content changes constantly, but most analysis techniques (starting next week: DFT/FFT) assume the signal is stationary over the window they look at.
- Solution: **framing** — cut the signal into short, overlapping windows (~25 ms, sliding by ~10 ms), short enough to look roughly stationary inside each one.
- Each frame is multiplied by a **window** (e.g., Hamming) that fades its edges toward zero; overlap gives samples near one frame's edge full weight in the next frame.



Tip
"Frame length and hop size" reappear in exactly this form for STFT and MFCCs next week.

Part 4

Describing What We Hear

Pitch, loudness, intensity, F0

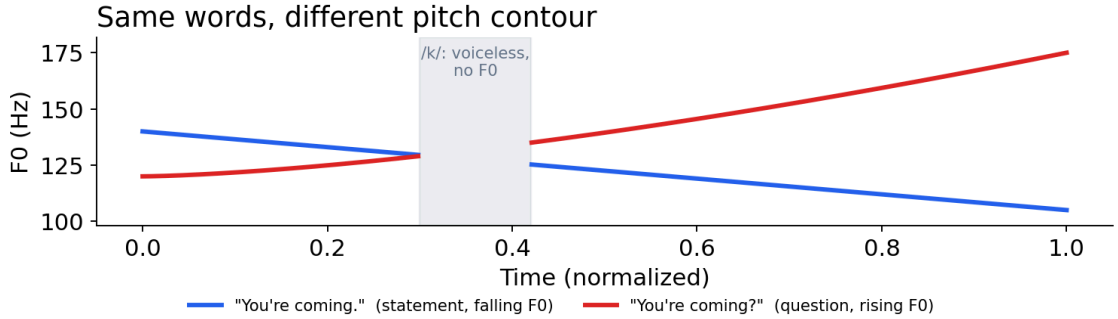
Pitch, Loudness, Intensity

- **Intensity**: the power per unit area carried by the sound wave, proportional to amplitude squared ($I \propto A^2$), in decibels relative to a reference (dB SPL in air).
- **Loudness**: our *perception* of intensity. Related to intensity, but not identical — perception is nonlinear and frequency-dependent.
- **Pitch**: our *perception* of frequency — how high or low a sound seems. Related to frequency, but subjective.
- **F0 (fundamental frequency)**: the physical measurement most closely underlying perceived pitch: the repetition rate of a periodic waveform ($1/T$), i.e., the rate the vocal folds vibrate at.

> **Note**
> **F0/intensity** are what we measure from the signal; **pitch/loudness** are what a listener experiences. They correlate strongly but are not the same thing.

F0 Over Time: the Pitch Contour

- F0 isn’t a single number for a whole utterance — it changes continuously. Tracking it frame-by-frame gives a **pitch contour** (or **pitch track**).
- The same words, said with a different pitch contour, can mean something different.
- F0 exists only where the vocal folds vibrate: the track **breaks** at the voiceless /k/ of "coming". That gap is exactly the voiced/unvoiced marking you’ll do in Assignment 2.



> **Note**
> **RMS (root-mean-square) amplitude** — the standard frame-level measurement of intensity, computed the same frame-by-frame way as F0.

Part 5

Why Vowels Sound Different

The source-filter model, F1 and F2

The Source–Filter Model

- **Source**: in voiced sounds, the vocal folds vibrate, producing a buzz-like sound rich in harmonics (multiples of F0). In voiceless sounds, the source is turbulent **noise** at a constriction (/s/, /f/, aspiration, bursts), with no F0 at all. Voiced fricatives (/z/, /v/) use both sources at once.
- **Filter**: the vocal tract (throat, mouth, nose) is a resonant cavity whose *shape* — set by tongue position, jaw, lips — amplifies some frequencies and damps others.
- Change the vocal tract shape (move your tongue) without changing the source (F0), and you change which frequencies come through loudest — that's what turns the same buzz into different vowels.
- The frequencies the vocal tract resonates at, and therefore amplifies, are called **formants**.

> **Tip**
> Say "ee" then "ah" while gently touching your throat — the buzz (source) barely changes, but the sound changes completely. That's the filter doing the work.

What F1 and F2 Actually Are

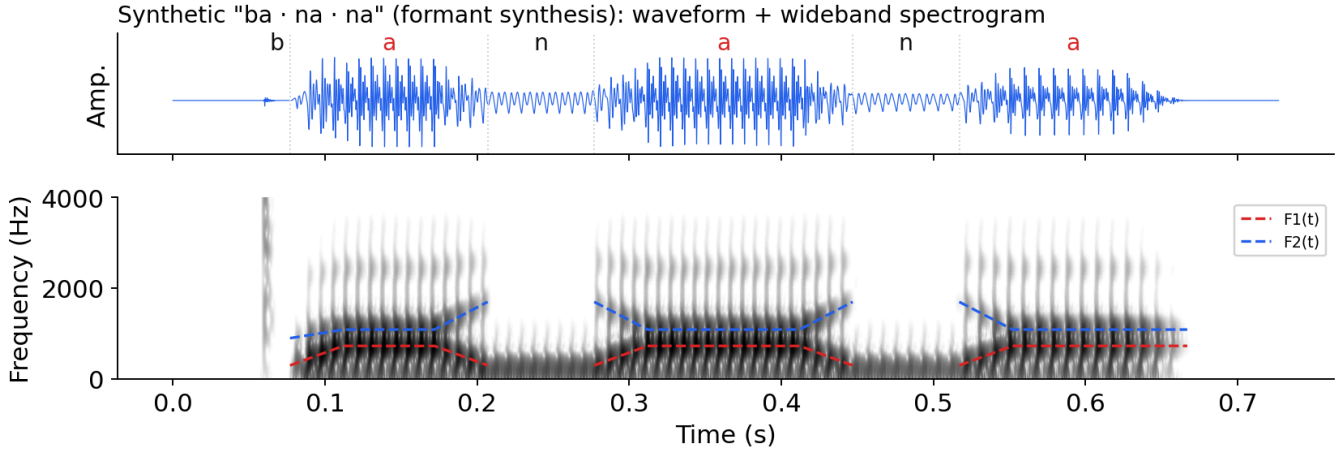
- **F1** is the vocal tract's *lowest* resonance, **F2** the *next* one up; they carry most of what distinguishes vowels.
- **F1 ↔ tongue height**: high ("ee") → low F1; low ("ah") → high F1.
- **F2 ↔ frontness**: front ("ee") → high F2; back ("oo") → low F2. **Lip rounding** also lowers F2.

Part 6

Putting It All Together

The spectrogram

The Spectrogram



- A **spectrogram** is a picture of sound: time on the x-axis, frequency on the y-axis, darkness showing energy (Praat's grayscale; in color maps such as magma, *brighter* = more energy).
- **Vowels**: vertical lines (one for each vocal cord vibration) with thick dark horizontal stripes (**formant**) showing the main pitch levels (F1 low, F2 above). **Nasal** still voiced but weaker, energy mostly in a low band near 250 Hz, with no burst or silence. **Bursts**: a thin vertical smudge (the /b/ here is short and weak). **Silence**: blank space.

Part 7
Conclude

What to Take Away

- A syllable is onset + nucleus (usually a vowel) + coda; vowels are open and normally voiced, consonants involve some constriction.
- Waveform = pressure over time; amplitude = size of the swing; frequency (Hz) = cycles/second; period = 1/frequency. Voiced sounds are quasi-periodic; voiceless sounds are noise.
- **Sampling** discretizes time: f_s must exceed twice the highest frequency (Nyquist); 16 kHz is the ASR standard. **Quantization** discretizes amplitude (16-bit standard). **Framing** cuts the signal into short, near-stationary windows.
- Intensity/F0 are physical measurements; loudness/pitch are what we perceive.
- **F1** mostly tracks tongue height, **F2** frontness (and lip rounding); vocal tract length shifts them all. Together they place a vowel in the vowel space, all visible at once in a **spectrogram**.

Next week: how a spectrogram is actually computed — DFT, FFT, windowing, and the mel scale.

Assignment 2: Due Next Week

Go back to **your own Assignment 1 recording** (the sentence you recorded and computed WER on).

1. **Plot it**: Mark voiced vs. unvoiced regions.
2. **Measure formants**: pick 2–3 vowels, measure F1/F2 with Praat; librosa has no formant tracker.
3. **Annotate a spectrogram** by hand — mark formant bands, bursts, and silence.
4. **Degrade it**, re-run each version through your Assignment 1 ASR system, and recompute WER. First resample your recording (probably 44.1/48 kHz) to 16 kHz.
   - **Downsample** 16k → 8k → 4k.
   - **Requantize** 16 → 8 → 4 bit.
5. **Report**: plot WER vs. sampling rate / bit depth. Where does degradation become *audible*, and where does the ASR system actually *break*? Do those points coincide?

Learn More

Course Homepage · J&M Ch. 16 · WASP · Chrome Music Lab: Spectrogram